

MD ALOMGEER HUSSEIN

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SUMMARY

Ph.D. researcher in Information Systems specializing in Large Language Models, NLP, and human-centered AI, with a focus on patient-facing AI in high-stakes settings. Build and fine-tune LLMs (LoRA/PEFT, quantization), design LLM evaluation, safety, and RAG pipelines, and analyze large real-world datasets comparing human and LLM responses on safety, empathy, personalization, and trust. First-place finisher in an NLM/NIH-organized question-answering shared task, with peer-reviewed publications and grant-funded projects.

TECHNICAL SKILLS

Languages:	Python, R, SQL, MATLAB
ML/DL Frameworks:	PyTorch, TensorFlow, Keras, HuggingFace Transformers, scikit-learn, OpenCV
Data & Tooling:	NumPy, Pandas, Git, Docker, Linux
ML/AI Methods:	LLM Fine-tuning (LoRA/PEFT), RLHF, Quantization, RAG, Agentic AI, LLM & Patient-Centered AI Evaluation, Safety & Empathy Evaluation, Grounded QA, Transformers, NLP, Human-centered AI, Explainable AI, Causal Inference
Cloud & MLOps:	AWS, Google Cloud Platform, Vertex AI

SELECTED PROJECTS

Cascaded LLM Pipeline for Grounded Clinical QA (ArchEHR-QA 2026 Shared Task) 2026

- Built a cascaded LLM pipeline for grounded clinical question answering as part of the HealthNLP_Retrievers team.
- **Won 1st place** in Subtask 1 (Question Interpretation) at the NLM/NIH-organized CL4Health @ LREC 2026 shared task; also ranked 5th (Answer Generation), 7th (Evidence Identification), and 9th (Answer-Evidence Alignment).

Patient-Centered Evaluation of LLM Responses to Patient Questions 2025 – 2026

- Designed a multi-dimensional evaluation of LLM-generated answers to patients' lab-test questions across readability, completeness, personalization, transparency, and emotional support.
- Benchmarked LLM responses against human responses from an online health community. (Accepted, AMIA 2026.)

QnA: LLM-Powered Conversational Agent for Colorectal Cancer Knowledge Fall 2024

- Fine-tuned Meta LLaMA-2 7B Chat on 1,000,000+ online health-community posts/comments to deliver accurate, empathetic domain-specific answers.
- Applied 4-bit quantization and LoRA fine-tuning to cut compute cost while adapting the model to the clinical domain.

TriAudit: Auditing and Improving Trustworthy LLMs for Health Consumers 2026

- Developing an auditing framework to measure and improve the trustworthiness of consumer-facing health LLMs. (Under review, ICMLA 2026.)

From Data to Topography: Deep Learning for Ice-Bed Prediction Spring 2023

- Built a dense + LSTM model to predict Greenland's ice-bed topography from sparse satellite data, outperforming traditional ML baselines.

Detecting Fraudulent Job Postings with NLP Fall 2022

- Built an NLP classifier over a 17,880-record Kaggle dataset using Multinomial Naive Bayes, KNN, and SVM to flag deceptive job ads.

EXPERIENCE

Graduate Research Assistant — Care Research Lab, UMBC Spring 2025 – Present

- Lead ML/NLP research on a Google Society-Centered AI grant building a participatory AI framework that helps underserved patients understand and use their medical records.

- Contribute to an NIH/NIMHD-funded social engagement support system, evaluating outcomes across demographic and geographic subgroups.

Graduate Teaching Assistant — Department of Information Systems, UMBC Spring 2023 – Fall 2024

- Taught and supported courses of 36–200 students in databases, systems, and management science; led grading, course development, and tutoring.

EDUCATION

Ph.D. in Information Systems , University of Maryland, Baltimore County	August 2022 – Present
M.Sc. in Information Systems , University of Maryland, Baltimore County	August 2022 – December 2024 (GPA: 3.84/4.0)
M.Sc. in Electrical and Electronic Engineering , University of Dhaka, Bangladesh	December 2015 – October 2017
B.Sc. in Electrical and Electronic Engineering , University of Dhaka, Bangladesh	January 2011 – December 2015

SELECTED PUBLICATIONS

- Hussein MA, Doshi R, He L, Reynolds TL. Comparing Human and Large Language Model Responses to Patients’ Online Questions: Towards Multi-dimensional Patient-centered Support. *AMIA Annual Symposium Proceedings*; 2026. (Accepted)
- HealthNLP_Retrievers at ArchEHR-QA 2026: Cascaded LLM Pipeline for Grounded Clinical Question Answering. *CL4Health @ LREC 2026 Shared Task*; 2026. (Accepted)
- Hosen MB, Rupa AP, Hussein MA. Beyond the Visual Shortcut: Confound-Controlled Multimodal Drug-Sale Intent Detection on Instagram. *ICMLA*; 2026. (Under Review)

E: authors . From Code to Career: Assessing Competitive Programmers for Industry Placement. *IEEE QPAIN*; 2025.

AWARDS & HONORS

• COEIT Research Day Student Award, UMBC	2026
• 1st Place, Subtask 1 (Question Interpretation), ArchEHR-QA 2026 Shared Task (NLM/NIH-organized)	2026
• Honorable Mention, COEIT Research Day Student Award, UMBC	2025